

Advanced Machine Learning

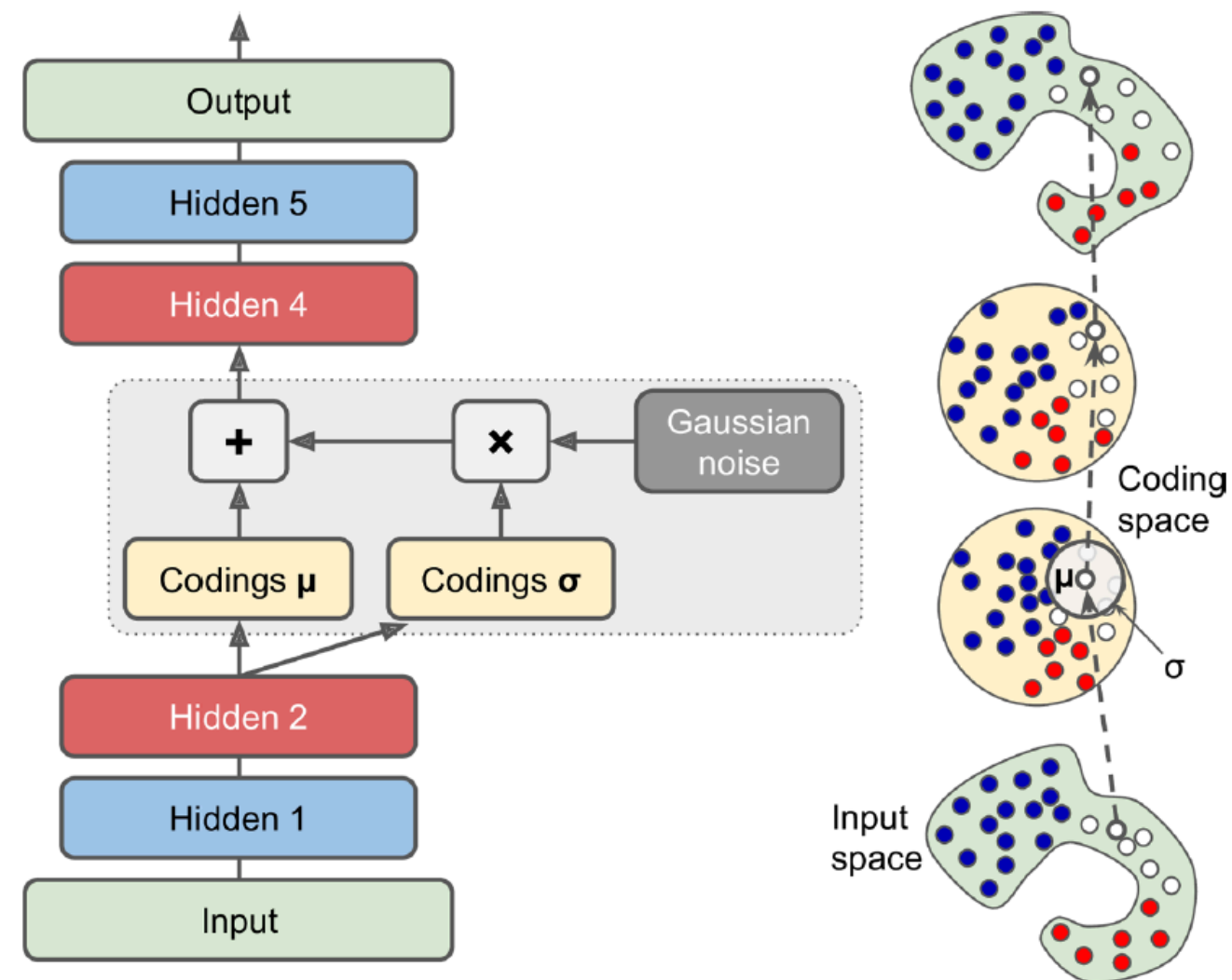
DATA 642

Lecture 14 — Variational Autoencoders

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Variational Autoencoder

- Variational autoencoders are probabilistic autoencoders, meaning that their outputs are partly determined by chance, even after training (as opposed to denoising autoencoders, which use randomness only during training).
- Most importantly, they are generative autoencoders, meaning that they can generate new instances that look like they were sampled from the training set.

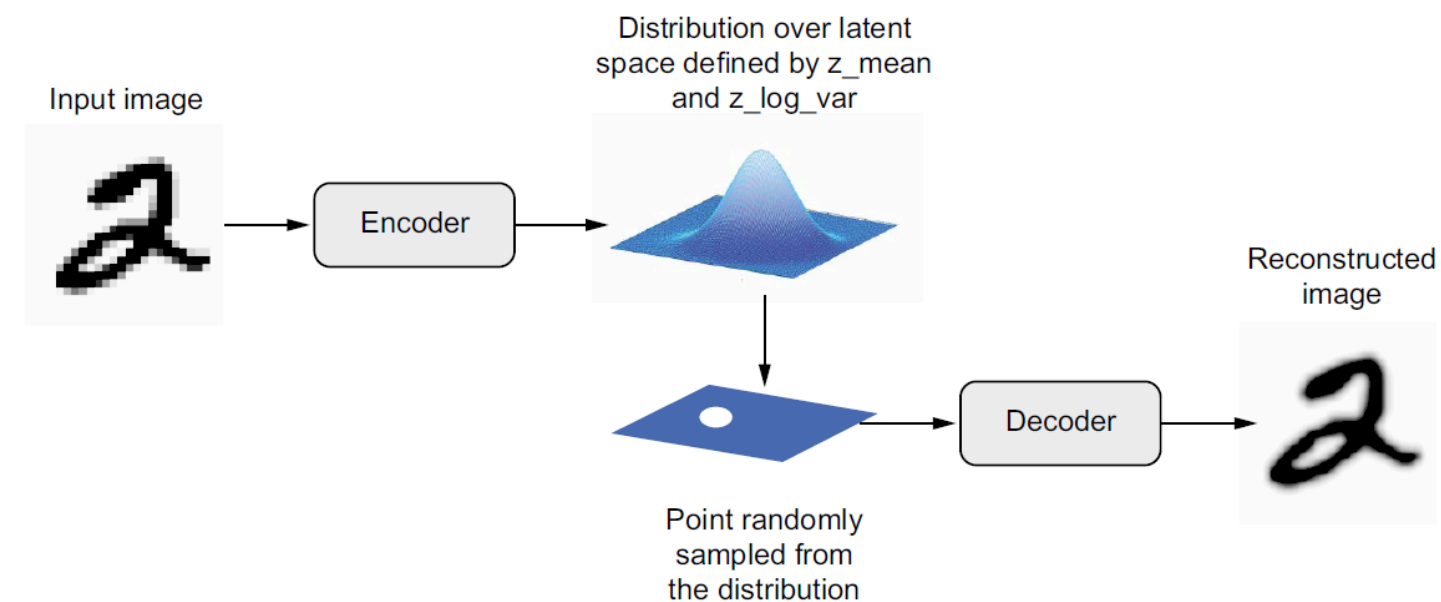


Applications and Use Cases of VAEs

1. **Image Generation:** VAEs are widely used in generating realistic images, such as in the creation of art, digital avatars, and synthetic imagery for gaming and entertainment industries.
2. **Anomaly Detection:** VAEs can be utilized for anomaly detection in various systems, such as detecting fraudulent transactions in finance, identifying defects in manufacturing processes, or spotting anomalies in medical images for disease diagnosis.
3. **Data Compression:** VAEs can compress high-dimensional data into lower-dimensional representations while retaining essential information. This is beneficial for efficient storage, transmission, and processing of large datasets, particularly in fields like telecommunications and multimedia.
4. **Drug Discovery:** VAEs are employed in drug discovery and molecular design tasks, where they generate novel chemical compounds with desired properties, accelerate the process of drug development, and optimize molecular structures.
5. **Natural Language Processing (NLP):** VAEs play a crucial role in generating coherent and contextually relevant text, such as in chatbots, language translation, and content generation for digital marketing.

The procedure

- Instead of directly producing a coding for a given input, the encoder produces a mean coding μ and a standard deviation σ . The actual coding is then sampled randomly from a Gaussian distribution with mean μ and a standard deviation σ . After that the decoder decodes the sampled coding normally.
- During training, the cost function pushes the codings to gradually migrate within the coding space (also called the latent space) to end up looking like a cloud of Gaussian points.
- One great consequence is that after training a variational autoencoder, you can very easily generate a new instance: just sample a random coding from the Gaussian distribution, decode it, and here you are!



The cost function

- It is composed of two parts. The first is the usual reconstruction loss that pushes the autoencoder to reproduce its inputs (we can use cross entropy for example or the negative log-likelihood of the data under a Gaussian distribution).
- The second is the latent loss that pushes the autoencoder to have codings that look as though they were sampled from a simple Gaussian distribution: it is the KL divergence between the target distribution (i.e., the Gaussian distribution) and the actual distribution of the codings.

The math details of the cost function

The objective of a VAE is to maximize the variational lower bound, also known as the evidence lower bound (ELBO). The ELBO is derived from the joint distribution of the data and latent variables, and it serves as a tractable approximation to the true log likelihood of the data.

Let $\mathbf{x}$ denote the observed data, $\mathbf{z}$ denote the latent variables, $\boldsymbol{\theta}$ denote the parameters of the decoder network, and $\boldsymbol{\phi}$ denote the parameters of the encoder network.

The ELBO, denoted by $\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x})$, is given by:

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}) = \mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) \quad (1)$$

where $p(\mathbf{x}|\mathbf{z})$ is the likelihood of the data given the latent variables, $q(\mathbf{z}|\mathbf{x})$ is the approximate posterior distribution of the latent variables given the data, and $p(\mathbf{z})$ is the prior distribution of the latent variables.

The objective is to maximize the ELBO with respect to the parameters θ and ϕ , which can be written as:

$$\max_{\theta, \phi} \mathcal{L}(\theta, \phi; x) \tag{2}$$

This maximization is typically achieved through gradient-based optimization methods such as stochastic gradient descent (SGD) or Adam.

By maximizing the ELBO, the VAE learns to reconstruct the input data while simultaneously regularizing the latent space to follow a prior distribution. This encourages the VAE to learn meaningful representations of the data in the latent space.

A simplified cost function under the Gaussian assumption

Using the Gaussian assumption, the reconstruction term $\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z})]$ can be written as the negative mean squared error between the input data $\mathbf{x}$ and the reconstructed output $\hat{\mathbf{x}}$:

$$\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z})] = -\frac{1}{2} \sum_{i=1}^N \left(\frac{(x_i - \hat{x}_i)^2}{\sigma_i^2} + \log(\sigma_i^2) + \log(2\pi) \right)$$

where N is the dimensionality of the data, x_i is the i -th element of the input data $\mathbf{x}$, $\hat{x}_i$ is the i -th element of the reconstructed output $\hat{\mathbf{x}}$, and σ_i^2 is the variance of the i -th dimension of the data.

The KL divergence term $D_{\text{KL}}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$ between the approximate posterior and the prior can be expressed in closed form due to the Gaussian assumption:

$$D_{\text{KL}}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) = -\frac{1}{2} \sum_{i=1}^N (1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2)$$

where μ_i is the mean of the i -th dimension of the latent variables $\mathbf{z}$.

Therefore, the complete ELBO expression for a VAE under the Gaussian assumption is given by the combination of the reconstruction term and the KL divergence term:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \boldsymbol{x}) = & -\frac{1}{2} \sum_{i=1}^N \left(\frac{(x_i - \hat{x}_i)^2}{\sigma_i^2} + \log(\sigma_i^2) + \log(2\pi) \right) \\ & - \frac{1}{2} \sum_{i=1}^N (1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2)\end{aligned}$$

This expression represents the ELBO for a VAE assuming Gaussian distributions for both the approximate posterior and the prior in the latent space.